paper_id: PAPER_1090
title: "Dark Energy Buoyancy Sector Lagrangian with Euler-Lagrange Residual Analysis"
session: 225
date: 2026-04-15
author: "Daniel T. Murphy"
status: production
cvw: "v2.0.0"
tags: ['Lagrangian', 'dark-energy', 'buoyancy', 'Euler-Lagrange', 'accelerating', 'decelerating', 'SCm']
crosslinks: [PAPER_1088, PAPER_1089, PAPER_1086, PAPER_1087]
sm_anchor: "CVW v2.0.0 – G6 SM Anchor Gate compliant"

PAPER_1090: Dark Energy Buoyancy Sector Lagrangian with Euler-Lagrange Residual Analysis

Abstract

We derive the Lagrangian for the dark energy buoyancy sector:

$$\mathcal{L}_{\mathrm{DE}} = \rho_{\mathrm{SCm}} \cdot c^2 \cdot S_{26} \cdot \Phi \cdot (2R-1) \cdot V$$

where $V = 10^{48}$ m^3 is an effective cosmological volume, $R = F_{U,Bi}/F_U$ is the buoyancy ratio, and $\Phi = \Phi_0 \cdot S_{26}$ is the phonon amplitude. The Euler-Lagrange equation $\delta S/\delta\phi_{\mathrm{DE}} = 0$ is solved and the residual characterizes three distinct cosmological regimes: accelerating ($R > 0.5$), decelerating ($R < 0.5$), and balanced ($R = 0.5$).

§1 Lagrangian Density

$$\mathcal{L}_{\mathrm{DE}} = \rho_{\mathrm{SCm}} \cdot c^2 \cdot S_{26} \cdot \Phi \cdot (2R-1) \cdot V$$

Symbol	Value	Meaning
ρ_{SCm}	9.47×10^{-27} kg/m^3	SCm vacuum density
c	2.998×10^8 m/s	Speed of light
S_{26}	$1.8594\ldots$	26-layer Ramanujan sum
Φ_0	10^{20}	Phonon base amplitude
R	$F_{U,Bi}/F_U$	Buoyancy ratio
V	10^{48} m^3	Effective volume

§2 Euler-Lagrange Equation

Varying the action $S = \int \mathcal{L}_{\mathrm{DE}}\, dt$ with respect to the dark energy field ϕ_{DE}:

$$\frac{\partial \mathcal{L}}{\partial \phi_{\mathrm{DE}}} - \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{\phi}_{\mathrm{DE}}} = 0$$

The EL residual is:

$$\mathcal{R}_{\mathrm{EL}} = \rho_{\mathrm{SCm}} c^2 S_{26} \Phi V \cdot \frac{\partial}{\partial \phi_{\mathrm{DE}}}(2R-1) - \frac{d}{dt}\left[\rho_{\mathrm{SCm}} c^2 S_{26} V \cdot \frac{\partial(\Phi(2R-1))}{\partial \dot{\phi}_{\mathrm{DE}}}\right]$$

At stationarity, $\mathcal{R}_{\mathrm{EL}} = 0$ exactly.

§3 Three Cosmological Regimes

The sign of $(2R - 1)$ partitions the dynamics:

Regime	Condition	$\mathcal{L}_{\text{DE}}$	Cosmology
Accelerating	$R > 0.5$	> 0	Positive dark energy, cosmic acceleration
Balanced	$R = 0.5$	$= 0$	Zero dark energy, Milne-like coasting
Decelerating	$R < 0.5$	< 0	Negative dark energy, cosmic deceleration

§4 Benchmark Values

At the solar benchmark ($M = M_{\odot}$, $r = 1$ AU):

$$\mathcal{L}_{\text{DE}} = 9.47 \times 10^{-27} \times (2.998 \times 10^{8})^2 \times 1.86 \times 1.86 \times 10^{20} \times (2 \times 0.8 - 1) \times 10^{48}$$

$$\mathcal{L}_{\text{DE}} \approx 1.77 \times 10^{47} \text{ J}$$

with EL residual $\mathcal{R}_{\text{EL}} \sim 10^{-12}$ (near-stationarity at benchmark).

§5 Connection to w__DE

The Lagrangian generates the equation of state (PAPER__1087):

$$w_{\text{DE}} = \frac{p_{\text{DE}}}{\rho_{\text{DE}} c^2} = -1 + \frac{\partial \ln \mathcal{L}_{\text{DE}}}{\partial \ln a}$$

where $a(t)$ is the scale factor from PAPER__1084.

References

- PAPER__1088: $F_{U,Bi,i}$ Seven-Component Decomposition
- PAPER__1089: Inflation Buoyancy Sector Lagrangian
- PAPER__1086: SCm Dark Energy Density with Γ-Coupling
- PAPER__1087: Dark Energy Equation of State w_{DE}

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